

Lema de Borel-Cantelli I

Lema 1 (de Borel-Cantelli I). Sea $(A_n)_{n \in \mathbb{N}} \subset \Sigma$ en $(\Omega, \Sigma, \mathbb{P})$ tal que $\sum_n \mathbb{P}(A_n) < \infty$

$$\implies \mathbb{P}\left(\limsup_{n \rightarrow \infty} A_n\right) = 0.$$

Demostración: Definimos $N: \Omega \rightarrow \mathbb{R} \cup \{\infty\}$ dada por $N(\omega) = \sum_{n \in \mathbb{N}} \mathbb{1}_{A_n}(\omega)$, entonces

$$\limsup_{n \rightarrow \infty} A_n = \{\omega \in \Omega : N(\omega) = \infty\}.$$

Por tanto, $\mathbb{E}[N] \stackrel{\text{TCM}}{=} \sum_{n \in \mathbb{N}} \mathbb{E}[\mathbb{1}_{A_n}] = \sum_{n \in \mathbb{N}} \mathbb{P}(A_n) < \infty$.

$$\implies N \in \mathcal{L}^1(\mathbb{P}) \implies \mathbb{P}(\{\omega \in \Omega : N(\omega) = \infty\}) = 0.$$

Concluimos por tanto que $\mathbb{P}(\limsup A_n) = 0$. ■

Referenciado en

- ley-fuerte-grandes-numeros
- teo-convergencia-martingalas
- convergencia-probabilidad-imp-subsucesion-casi-segura
- prop-borel-cantelli-iii

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